# Rijad Alisic

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## Education

2018–2023 *Doctor of Philosophy & Licentiate of Engineering,*
*KTH Royal Institute of Technology*
*Ph.D. thesis*: Defense of Cyber-Physical Systems Against Learning-Based Attackers, 2023
*Lic. thesis*: Privacy of Sudden Events in Cyber-Physical Systems, 2021
*Advisor*: Prof. Henrik Sandberg, *Co-advisor*: Prof. Karl Henrik Johansson

2013–2018 *Master of Science in Engineering Physics,*
*Faculty of Engineering LTH, Lund University*
*Grade average*: 5/5
*Specializations*: Automatic Control and Theoretical Physics
*Master thesis*: Distributed Heating Networks, 2018
*Advisor*: Assoc. Prof. Richard Pates, *Examiner*: Prof. Anders Rantzer

## Positions

2025–present *Wallenberg Postdoctoral Fellow,*
*Laboratory for Information and Decision Systems, Massachusetts Institute of Technology*
*Advisor*: Prof. Saurabh Amin
*Affiliations*: Lead Cybersecurity Researcher, MIT Maritime Consortium

2024–2025 *Postdoctoral Researcher, Wallenberg AI, Autonomous Systems and Software Program (WASP),*
*Division of Decision and Control Systems, KTH Royal Institute of Technology*
*Advisor*: Prof. Henrik Sandberg

2016–2019 *Mathematician, Department of Experimental Medical Science, Lund University*
A Multi-object Tracker for Social Interactions of Zebrafish Larvae, with Docent Fredrik Ek and Prof. Roger Olsson

2017 *Summer Undergraduate Research Fellow (SURF), California Institute of Technology*
A Controller Testbed using Temporal Logic, with Dr. Sumanth Dathathri and Prof. Richard Murray

## Publications

### Journal papers

- **R. Alisic**, J. Kim, and H. Sandberg, "Anomaly Detection with LWE Encrypted Control," *IEEE Transactions on Automatic Control*, conditionally accepted, 2025. arXiv:2502.10283.
- **R. Alisic**, P. E. Paré, and H. Sandberg, "Change Time Estimation Uncertainty in Nonlinear Dynamical Systems with Applications to COVID-19," *International Journal of Robust and Nonlinear Control*, 33(9):4732–4760, 2023.
- **R. Alisic**, J. Kim, and H. Sandberg, "Model-Free Undetectable Attacks on Linear Systems Using LWE-Based Encryption," *IEEE Control Systems Letters*, vol. 7, pp. 1249–1254, 2023.

### Conference papers

- J. Porzio, **R. Alisic**, and H. Sandberg, "Goodbye Innovation: Stealthy Destabilizing Sensor Attacks on Feedback Control Systems," to be presented at the *65th IEEE Conference on Decision and Control (CDC)*, 2026.
- **R. Alisic** and S. Amin, "How Sensor Attacks Transfer Across Lie Groups," to be presented at the *65th IEEE Conference on Decision and Control (CDC)*, 2026. arXiv:2604.03461.
- J. Porzio, A. Kanellopoulos, **R. Alisic**, G. Dán, and H. Sandberg, "SEC-Two: Secure Estimation over Two Channels," *American Control Conference (ACC)*, 2026.
- **R. Alisic** and H. Sandberg, "Privacy Enhancement of Structured Inputs in Cyber-Physical Systems,"

*60th IEEE Conference on Decision and Control (CDC)*, Austin, TX, 2021, pp. 4888–4894.

- **R. Alisic** and H. Sandberg, "Data-Injection Attacks Using Historical Inputs and Outputs," *European Control Conference (ECC)*, Delft, 2021, pp. 1399–1405.
- **R. Alisic**, M. Molinari, P. E. Paré, and H. Sandberg, "Ensuring Privacy of Occupancy Changes in Smart Buildings," *IEEE Conference on Control Technology and Applications (CCTA)*, Montreal, 2020, pp. 871–876.
- **R. Alisic**, M. Molinari, P. E. Paré, and H. Sandberg, "Maximizing Privacy in MIMO Cyber-Physical Systems Using the Chapman-Robbins Bound," *59th IEEE Conference on Decision and Control (CDC)*, Jeju, 2020, pp. 6272–6277.
- **R. Alisic**, P. E. Paré, and H. Sandberg, "Modeling and Stability of Prosumer Heat Networks," *IFAC-PapersOnLine*, vol. 52, no. 20, 2019, pp. 235–240.

## Awards

2024 *Top Downloaded Article 2022–2023, International Journal of Robust and Nonlinear Control, Wiley*
2020 *Student Travel Award, IEEE Control Systems Society*

## Awarded Stipends

2025 *Knut and Alice Wallenberg Foundation, stipend for postdoctoral studies at MIT, Boston, MA*
2022 *KTH Technical, Scientific, Research, and Education Fund, travel stipend for research exchange at the University of Melbourne*
2017 *Summer Undergraduate Research Fellowship stipend, California Institute of Technology*
2017 *Carl Erik Levin's Fund; Academic Society's Scholarship Fund; Sparbanken Syd's Scholarship Fund*
2016 *Archbishop H. Reuterdahl's Scholarship Fund; A.V. Bäcklund's Scholarship Fund*
2016 *Summer Scholarship for Undergraduate Research Project, Lund University*
2013 *Torsten Tegstam's Scholarship Fund, Sydskånska Gymnasieförbundet*

## Teaching and Supervision

2026 *Supervisor, MSc Thesis, KTH Royal Institute of Technology*
"Cyber-Physical Security in Direct Data-Driven Control Systems," Gustaf Zachrisson. Co-supervisors: Alf Isaksson (ABB), Sribalaji Coimbatore Anand (KTH). Examiner: Henrik Sandberg.

2026 *Supervisor, MSc Thesis, Massachusetts Institute of Technology*
"Anomaly Detection for GPS Spoofing in Maritime Cyber-Physical Systems: Filter-Based and Deep Learning Methods," Strahinja Janjusevic. With Themistoklis Sapsis, Saurabh Amin, Fotini Christia.

2025 *Supervisor, MSc Thesis, KTH Royal Institute of Technology*
"Automatic PID Tuning for Wastewater Pump Station: Run Pumps Efficiently With the Press of a Button," Balder Jörsäter. Co-advisor: Victor Molnö.

2025 *Supervisor, Summer Research Internship, Digital Futures, KTH*
"Synthetic Data Generation for Anomaly Detection," Johan Slettengren. With Aris Kanellopoulos and Henrik Sandberg.

2024 *Supervisor, Summer Research Internship, KTH & University of California, Berkeley*
"Can Transformers Learn Kalman Filters?" Wentinn Liao. With Christos Mavridis, Henrik Sandberg, Anant Sahai.

2021–2022 *Teaching Assistant, Advanced Control Theory, KTH*
Two times.

2021 *Supervisor, Summer Research Internship, Digital Futures, KTH*
"Security of Smart Home Sensors," Giorgio Sacchi.

2021 *Supervisor, MSc Thesis, KTH*
"Modeling and Evaluation of Turret Control Systems for Main Battle Tanks," Mikael Lyth.

2019 *Supervisor, BSc Thesis, KTH*
"Distributed Optimisation in Multi-Agent Systems Through Deep Reinforcement Learning," Andreas Eriksson & Jonas Hansson. Co-supervisor: Mina Ferizbegovic.

2018–2020 *Teaching Assistant, Basic Control Theory, KTH Royal Institute of Technology*
Three times.

2017–2018 *Teaching Assistant, Basic Control Theory, Lund University*
Three times.

## Research Visits

2024 *Seoul National University and Seoul National University of Science and Technology (SeoulTech)*
2024 *Excellence Center at Linköping–Lund in Information Technology (ELLIIT), Lund University*
2022 *University of Melbourne*

## Invited Talks

2026 *Invited Speaker, Workshop WS-08, IFAC World Congress, Busan, South Korea*
2025 *Invited Talk, Division of Systems and Control, Uppsala University*
2024 *Invited Speaker, ELLIIT Symposium, Lund University*
2024 *Invited Speaker, American Control Conference Student Workshop, Toronto*
2024 *Invited Lectures, SeoulTech, Seoul, South Korea*
2024 *Invited Seminar, Seoul National University, Seoul, South Korea*
2024 *Invited Talk, WASP Cybersecurity Cluster Meeting, Västervik*
2022 *Invited Seminar, Australian National University, Canberra*
2022 *Invited Seminar, University of Melbourne*

## Service

Fall 2026–present *Chair, LIDS Postdoc Committee, MIT*
2025 *Panelist, WARA Operational Data Workshop, Ericsson, Stockholm*
2024 *Lead, Postdoctoral Workplace Meeting (arbetsplatsträff), Division of Decision and Control Systems, KTH*
2024 *Chair of Mechatronics Session, European Control Conference, Stockholm*
2024 *Volunteer and Election Assistant, European Control Association; Volunteer, European Control Conference, Stockholm*
2020–2022 *Lab Responsible, Division of Decision and Control Systems, KTH*
2020 *Organizer, KTH IFAC 2020 Workshop and NetCon Workshop on Learning, Stockholm*
2018 *Volunteer, Her Tech Future, Automatic Control, Lund*

## Peer Review

Journals *IEEE Transactions on Automatic Control; Automatica; IEEE Transactions on Control of Network Systems; IEEE Transactions on Cyber-Physical Systems; IEEE Control Systems Letters; IEEE Transactions on Industrial Informatics; IET Cyber-Physical Systems: Theory & Applications; IEEE Access*
Conferences *IEEE Conference on Decision and Control (CDC); American Control Conference (ACC); European Control Conference (ECC); IEEE Conference on Control Technology and Applications (CCTA); IFAC World Congress*

## Languages and Skills

Languages: *Swedish* and *Bosnian* (native), *English* (fluent), *Spanish* (basic)
Advanced: *Python, Matlab, LaTeX*
Intermediate: *Java, ROS*
Basic: *JavaScript, Maple*